

AZRIN SULTANA

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EDUCATION

Master of Science (M.Sc.) in Computer Science | Jan 2025 – May 2026

American International University-Bangladesh (AIUB), Dhaka | CGPA: 3.91 / 4.0 | Magna Cum laude | Dean's award

Bachelor of Science (B.Sc.) in Computer Science and Engineering (CSE) | Jan 2018 – Sep 2021

American International University-Bangladesh (AIUB), Dhaka | CGPA: 3.92 / 4.0 | Magna Cum Laude | Dean's award

Higher Secondary School Certificate (HSC) | 2016

Kapasia College, Gazipur | GPA: 4.75 / 5.0

Secondary School Certificate (SSC) | 2024

Kapasia Pilot High School, Gazipur | GPA: 5.0 / 5.0 | Scholarship for results

INDUSTRY EXPERIENCE

AI Engineer | Aurlsoft | Remote | Jul 2025 – Feb 2026 (Contract)

- Designed and developed an AI-powered chatbot using large language models, retrieval-augmented generation (RAG), and agentic AI frameworks.
- Fine-tuned and optimized using QLoRA and LoRA techniques for generation tasks.
- Built intelligent document retrieval integrating LangChain and vector databases (ChromaDB).
- Scraped knowledge from various sources and annotated it.

Software Engineer | Aurlsoft | Remote | Oct 2021 – Sep 2024

- Designed and implemented responsive and interactive front-end interfaces and RESTful APIs using Node.js and Express.js for scalable backend services.
- Designed and managed MongoDB databases, including schema design and data optimization.
- Integrated third-party APIs and backend services to support application functionality.
- Implemented authentication and authorization systems using JWT and secure API practices.
- Debugged, tested, and optimized applications to improve performance and reliability.

TEACHING & ACADEMIC EXPERIENCE

Teaching Assistant (Computer Science) | American International University-Bangladesh (AIUB) | Dhaka, Bangladesh | Feb 2021 – May 2021

- Delivered lab sessions to 40+ undergraduate students in Computer Science and Engineering courses, covering programming, data structures, and algorithmic problem-solving.
- Designed and evaluated lab assessments, providing structured feedback to improve students' conceptual understanding and coding proficiency.
- Provided one-on-one academic counseling for students struggling with complex topics, including algorithms, database systems, and object-oriented programming.
- Collaborated with faculty to prepare course lecture slides, lab manuals, and supplementary teaching materials aligned with university curriculum standards.

RESEARCH PUBLICATIONS | 210+ CITATIONS | 4 PEER-REVIEWED JOURNALS | 3 PREPRINTS | 4 UNDER REVIEW

Peer-Reviewed Journal Articles:

- Islam, R., Sultana, A., & Tuhin, M. N. (2024). **A comparative analysis of machine learning algorithms with tree-structured Parzen estimator for liver disease prediction**. Elsevier, <https://doi.org/10.1016/j.health.2024.100358>
- Islam, R., Sultana, A., & Islam, M. R. (2024). **A comprehensive review for chronic disease prediction using machine learning algorithms**. Springer, <https://doi.org/10.1186/s43067-024-00150-4>
- Sultana, A., & Islam, R. (2023). **Machine learning framework with feature selection approaches for thyroid disease classification and associated risk factors identification**. Springer, <https://doi.org/10.1186/s43067-023-00101-5>

- Islam, R., Sultana, A., et al. (2023). **Clinical decision support system for diabetic patients by predicting Type 2 diabetes**. Wiley, <https://doi.org/10.1155/2023/6992441>

Under Review & Completed papers:

- Sultana, A., & Ahmed, F. (2025). **Explicit Grammar Semantic Feature Fusion for Robust Text Classification**. arXiv:2602.20749
- Sultana, A., & Ahmed, F. (2025). **A Lightweight Vision-Language Fusion Framework for Predicting App Ratings from User Interfaces and Metadata**. arXiv:2602.20531
- Sultana, A., & Chayon, H. R. (2025). **Entity-aware and secure query optimization in databases using named entity recognition**. arXiv:2601.01254
- Sultana, A., Mazid Ul Haque & Dip Nandi (2026). **An NLP-Driven Framework for Discovering Understudied Environmental Factors in Chronic Diseases from Medical Literature**.
- Sultana, A., & Dip Nandi (2026). **LiSTA-NER: A Lipschitz-Guided Learnable Activation Function for Stable Named Entity Recognition**.
- Sultana, A., & Khandaker Tabin Hasan (2026). **Large Language Models in the Software Development Lifecycle: A Comprehensive Systematic Review**
- Sultana, A., & Shamim Kaiser. (2026). **PHYLO-RAG: Gene Function Prediction in Organisms Using Homology-Aware Retrieval-Augmented Generation**

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, C#, JavaScript, PHP, SQL, HTML/CSS

AI & Machine Learning: Large Language Models, RAG, Agentic AI, QLoRA, LoRA, NLP, Deep learning, Computer vision

Libraries & Frameworks: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, LangChain, ReactJS, Django

Databases & Tools: MySQL, PostgreSQL, MongoDB, ChromaDB, Tableau

AWARDS & HONORS

Magna Cum Laude | AIUB | 2026 — Academic honors for M.Sc. in Computer Science (CGPA 3.91/4.0).

Dean's Award | AIUB | 2026 — Awarded for outstanding academic results for the academic year.

Magna Cum Laude | AIUB | 2021 — Academic honors for B.Sc. in CSE (CGPA 3.92/4.0).

Dean's Award | AIUB | 2021 — Awarded for outstanding academic results in the B.Sc. program.

CERTIFICATES

AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents  | Udemy | 2026

AI Engineer Production Track: Deploy LLMs & Agents at Scale | Udemy | 2026

RESEARCH INTERESTS

Large Language Models, RAG, Agentic AI, Trustworthy AI, Multimodal learning, Healthcare AI, Knowledge Graphs, Deep Learning for Clinical Decision Support.

REFERENCES

PROF. DR. DIP NANDI, Dean, Department of Computer Science and Engineering

American International University-Bangladesh (AIUB), Dhaka, Bangladesh

Email: dip.nandi@aiub.edu **Relation:** Academic Supervisor (M.Sc. Thesis Advisor)

DR. RAZUAN KARIM, Associate Professor, Department of Computer Science and Engineering

American International University-Bangladesh (AIUB), Dhaka, Bangladesh

Email: rkarim@aiub.edu **Phone:** +880 1741515573 **Relation:** Course Supervisor & Faculty Mentor

PROF. DR. FIROZ AHMED, Professor, Department of Computer Science and Engineering

American International University-Bangladesh (AIUB), Dhaka, Bangladesh

Email: fahmed@aiub.edu **Phone:** +880 1889402460 **Relation:** Course Supervisor & Research Collaborator